

Continuous Relaxation for Graph Isomorphism Detection

1 Problem Formulation

Given two simple, undirected graphs $G_1 = (V, E_1)$ and $G_2 = (V, E_2)$ on n vertices, we seek to determine whether they are isomorphic. Two graphs are isomorphic if there exists a permutation $\pi : V \rightarrow V$ such that $(u, v) \in E_1 \iff (\pi(u), \pi(v)) \in E_2$.

Let $\mathbf{A}, \mathbf{B} \in \{0, 1\}^{n \times n}$ denote the adjacency matrices of G_1 and G_2 respectively. A permutation π can be represented as a permutation matrix $\mathbf{P} \in \{0, 1\}^{n \times n}$ where $P_{ij} = 1$ iff $\pi(i) = j$. The graphs are isomorphic iff there exists a permutation matrix $\mathbf{P}$ such that:

$$\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{B} \quad (1)$$

Since searching over all $n!$ permutations is computationally intractable, we relax the problem to a continuous optimization over *doubly stochastic matrices*.

2 Doubly Stochastic Relaxation

A matrix $\mathbf{X} \in \mathbb{R}^{n \times n}$ is **doubly stochastic** if:

$$X_{ij} \geq 0, \quad \sum_{j=1}^n X_{ij} = 1 \quad \forall i, \quad \sum_{i=1}^n X_{ij} = 1 \quad \forall j \quad (2)$$

The set of doubly stochastic matrices forms the *Birkhoff polytope* $\mathcal{B}_n$, whose vertices are exactly the $n!$ permutation matrices (Birkhoff–von Neumann theorem). By optimizing over $\mathcal{B}_n$, we obtain a convex relaxation of the combinatorial problem.

3 Objective Function

We formulate a quadratic program (QP) that penalizes structural mismatch between the graphs under a soft permutation $\mathbf{X} \in \mathcal{B}_n$. The implemented objective combines four signals:

1. a degree-profile compatibility term,
2. a first-order adjacency commutator penalty,
3. a second-order (two-hop) commutator penalty,
4. a spectral alignment penalty based on Laplacian eigenvectors.

3.1 Degree Profile Compatibility (Linear Term)

Define the degree vectors

$$d_i^A = \sum_k A_{ik}, \quad d_j^B = \sum_k B_{jk}, \quad (3)$$

and the one-hop degree-profile summaries

$$s_i^A = \sum_k A_{ik} d_k^A, \quad s_j^B = \sum_k B_{jk} d_k^B. \quad (4)$$

The node-compatibility cost matrix is

$$C_{ij} = \alpha |d_i^A - d_j^B| + \beta |s_i^A - s_j^B|, \quad (5)$$

where $\alpha, \beta \geq 0$ are tunable weights. This penalizes mapping node i in G_1 to node j in G_2 when their local degree statistics differ. The linear objective term is:

$$\mathcal{L}(\mathbf{X}) = \sum_{i=1}^n \sum_{j=1}^n C_{ij} X_{ij} = \langle \mathbf{C}, \mathbf{X} \rangle_F \quad (6)$$

3.2 Adjacency Mismatch (Quadratic Term)

For a true permutation matrix $\mathbf{P}$, the transformed adjacency would be $\mathbf{P}^\top \mathbf{A} \mathbf{P}$. We penalize the discrepancy between $\mathbf{A}$ transformed by $\mathbf{X}$ and $\mathbf{B}$ transformed by $\mathbf{X}$:

$$\mathcal{Q}(\mathbf{X}) = \sum_{p=1}^n \sum_{q=1}^n \left(\sum_{k=1}^n A_{pk} X_{kq} - \sum_{k=1}^n X_{pk} B_{kq} \right)^2 \quad (7)$$

This can be written in matrix form. Let $\mathbf{R} = \mathbf{A} \mathbf{X} - \mathbf{X} \mathbf{B}$, then:

$$\mathcal{Q}(\mathbf{X}) = \|\mathbf{A} \mathbf{X} - \mathbf{X} \mathbf{B}\|_F^2 = \sum_{p,q} R_{pq}^2 \quad (8)$$

When $\mathbf{X} = \mathbf{P}$ is a valid isomorphism, we have $\mathbf{A} \mathbf{P} = \mathbf{P} \mathbf{B}$, hence $\mathcal{Q}(\mathbf{P}) = 0$.

3.3 Two-Hop Neighborhood Penalty

Direct adjacency alone does not fully capture local structure. Let

$$\mathbf{A}^{(2)} = \mathbf{A}^2, \quad \mathbf{B}^{(2)} = \mathbf{B}^2, \quad (9)$$

whose entries count length-2 walks and therefore encode common-neighbor structure. We add the penalty

$$\mathcal{T}(\mathbf{X}) = \|\mathbf{A}^{(2)} \mathbf{X} - \mathbf{X} \mathbf{B}^{(2)}\|_F^2. \quad (10)$$

As with the first-order commutator, if $\mathbf{X} = \mathbf{P}$ is an exact isomorphism then $\mathbf{A}^2 \mathbf{P} = \mathbf{P} \mathbf{B}^2$, so $\mathcal{T}(\mathbf{P}) = 0$.

3.4 Spectral Alignment Penalty

Let the combinatorial Laplacians be

$$\mathbf{L}_A = \text{diag}(\mathbf{A}\mathbf{1}) - \mathbf{A}, \quad \mathbf{L}_B = \text{diag}(\mathbf{B}\mathbf{1}) - \mathbf{B}. \quad (11)$$

Let $\mathbf{U}_A, \mathbf{U}_B \in \mathbb{R}^{n \times n}$ be matrices of Laplacian eigenvectors ordered by increasing eigenvalue. In implementation, each eigenvector column is given a deterministic sign convention to reduce the usual sign ambiguity.

We then penalize disagreement between the spectral embedding of G_1 and the transported embedding of G_2 :

$$\mathcal{S}(\mathbf{X}) = \|\mathbf{U}_A - \mathbf{X}\mathbf{U}_B\|_F^2. \quad (12)$$

This term gives the solver a global structural signal even when $\mathbf{X}$ is far from permutation-like.

4 Quadratic Program

The complete implemented optimization problem is:

$$\begin{aligned} Z^* = \min_{\mathbf{X}} \quad & \mathcal{L}(\mathbf{X}) + \mathcal{Q}(\mathbf{X}) + \lambda_T \mathcal{T}(\mathbf{X}) + \lambda_S \mathcal{S}(\mathbf{X}) \\ \text{s.t.} \quad & \sum_{j=1}^n X_{ij} = 1 \quad \forall i \in \{1, \dots, n\} \\ & \sum_{i=1}^n X_{ij} = 1 \quad \forall j \in \{1, \dots, n\} \\ & X_{ij} \geq 0 \quad \forall i, j \end{aligned} \quad (13)$$

Here $\lambda_T, \lambda_S \geq 0$ control the contribution of the two-hop and spectral penalties respectively. In the current codebase the default values are

$$\alpha = 1.0, \quad \beta = 0.35, \quad \lambda_T = 0.2, \quad \lambda_S = 0.15. \quad (14)$$

The optimization remains a quadratic program over the Birkhoff polytope and is solved numerically with Gurobi.

4.1 Remark on Entropy Regularization

An additional entropy term,

$$-\varepsilon \sum_{i,j} X_{ij} \log X_{ij}, \quad (15)$$

was proposed as a continuation-style regularizer to keep $\mathbf{X}$ diffuse early in the solve. This term is *not* currently implemented in the codebase, because it requires a different nonlinear or piecewise-linear treatment than the present quadratic formulation.

5 Isomorphism Index

The optimal objective value Z^* measures the “distance” from isomorphism:

- $Z^* = 0$: The graphs are isomorphic (a perfect permutation exists)

- $Z^* > 0$: Structural mismatch; larger values indicate greater dissimilarity

We define a normalized **Isomorphism Index** $I \in (0, 1]$:

$$\boxed{I = \exp(-\lambda \cdot Z^*)} \tag{16}$$

where $\lambda > 0$ is a scaling parameter (default $\lambda = 0.1$). This maps the unbounded objective to a bounded similarity score:

- $I \approx 1$: Graphs are (likely) isomorphic
- $I \ll 1$: Graphs are structurally dissimilar

6 Interpretation of the Solution

The optimal $\mathbf{X}^*$ is a doubly stochastic matrix representing a “soft” node correspondence:

- If $\mathbf{X}^*$ is a permutation matrix (binary), it directly encodes the isomorphism mapping
- If $\mathbf{X}^*$ has fractional entries, the relaxation is not tight—this typically occurs for non-isomorphic graphs or for ambiguous graph symmetries

7 Complexity

The QP has n^2 variables and $2n$ equality constraints. Modern interior-point solvers achieve polynomial complexity in practice, making this approach tractable for moderate graph sizes ($n \lesssim 100$).